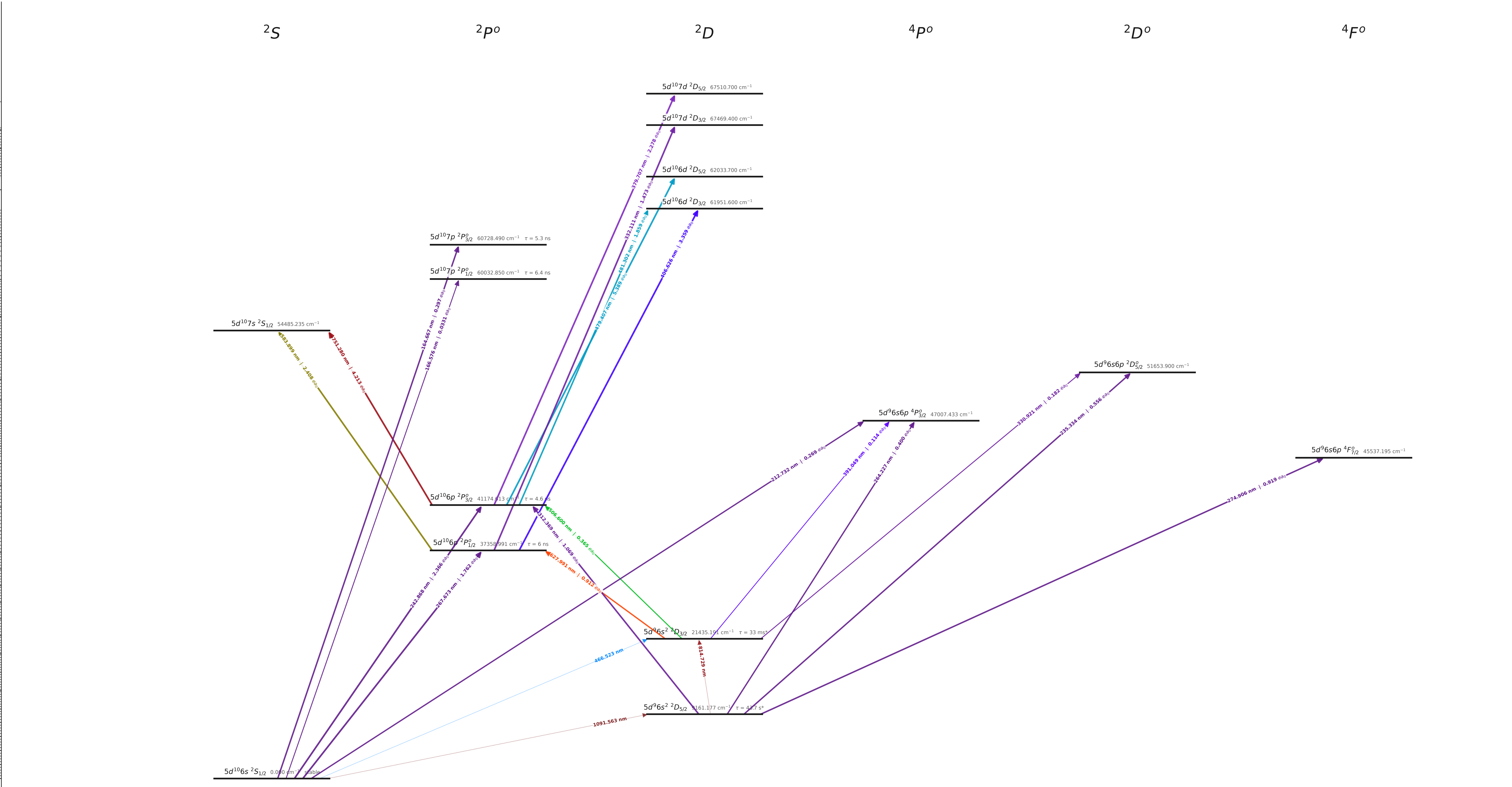


Gold

Au (Au I). Strongest classified transitions below 2 μm (NIST): 15 levels, 23 lines.

Au⁺ ionisation limit 74409.11 cm⁻¹ = 9.22555 eV (134.392 nm)

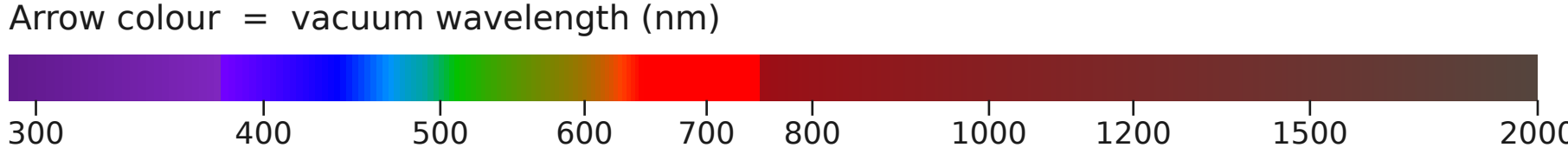


Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J') (ea₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ⁹ 6s ² ² D _{5/2} - 5d ⁹ 6s ² ² D _{3/2}	814.7294	814.5054	367.9657	-	2.990e+01	4.823e-06	98.7	main decay of 2D3/2
5d ¹⁰ 6s ² S _{1/2} - 5d ⁹ 6s ² ² D _{5/2}	1091.5628	1091.2639	274.6452	-	2.290e-02	3.642e-09	100	E2 clock transition proposed for a Au lattice clock

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |U||er||J') in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



197Au hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5d ¹⁰ 6s ² S _{1/2}	3049.660092(7)	-	F=1: -3812.07 F=2: +2287.24
5d ⁹ 6s ² D _{5/2}	80.236(3)	-1049.781(11)	F=1: -1156.09 F=2: -155.79 F=3: +557.32 F=4: +38.44
5d ⁹ 6s ² D _{3/2}	199.84250(20)	-911.0766(5)	F=0: -1888.26 F=1: -777.34 F=2: +533.43 F=3: +221.88
5d ¹⁰ 6p ² P° _{1/2}	321.7(12)	-	F=1: -402.12 F=2: +241.28
5d ¹⁰ 6p ² P° _{3/2}	14.0(5)	327.6(16)	F=0: +357.00 F=1: +43.40 F=2: -256.20 F=3: +113.40

Measured A, B constants; sources in the data file.